

## Solución Tercer Examen Parcial

1. Calcule las siguientes integrales:

a)  $\int x \sqrt{x+3} dx$

**3 Pts**

**Solución:**

$$u = x + 3 \Rightarrow u - 3 = x$$

$$du = dx$$

$$\int x \sqrt{x+3} dx$$

$$= \int (u - 3) \sqrt{u} du$$

$$= \int (u^{3/2} - 3u^{1/2}) du$$

$$= \frac{5u^{5/2}}{2} - \frac{3 \cdot 2u^{3/2}}{3} + C$$

$$= \frac{5(x+3)^{5/2}}{2} - 2(x+3)^{3/2} + C$$

b)  $\int y \sin(3y) dy$

**3 Pts**

**Solución:**

$$\begin{array}{l|l} u = y & dv = \sin(3y) dy \\ du = dy & v = -\frac{\cos(3y)}{3} \end{array}$$

$$\int y \sin(3y) dy = -\frac{y \cos(3y)}{3} + \frac{1}{3} \int \cos(3y) dy$$

$$= -\frac{y \cos(3y)}{3} + \frac{\sin(3y)}{9} + C$$

$$c) \int \frac{5x+11}{(x-2)(2x+3)} dx$$

4 Pts

**Solución:**

$$\frac{5x+11}{(x-2)(2x+3)} = \frac{A}{x-2} + \frac{B}{2x+3} = \frac{A(2x+3) + B(x-2)}{(x-2)(2x+3)}$$

$$\Rightarrow 5x+11 = A(2x+3) + B(x-2)$$

$$x=2 \Rightarrow 21 = 3A \Rightarrow A=3$$

$$x = -\frac{3}{2} \Rightarrow -\frac{15}{2} + 11 = B\left(-\frac{3}{2} - 2\right) \Rightarrow B = -1$$

$$\begin{aligned} & \int \frac{5x+11}{(x-2)(2x+3)} dx \\ &= \int \frac{3}{x-2} - \frac{1}{2x+3} dx \\ &= 3 \ln|x-2| - \frac{1}{2} \ln|2x+3| + C \end{aligned}$$

$$d) \int_1^{+\infty} x e^{-x^2} dx$$

4 Pts

**Solución:**

$$\begin{aligned} & \int_1^{+\infty} x e^{-x^2} dx \\ &= \lim_{p \rightarrow +\infty} \int_1^p x e^{-x^2} dx \end{aligned}$$

$$u = -x^2$$

$$du = -2x dx$$

$$x=1 \Rightarrow u=-1$$

$$x=p \Rightarrow u=-p^2$$

$$\begin{aligned} & \lim_{p \rightarrow +\infty} \int_{-1}^{-p^2} -\frac{e^u}{2} du \\ &= \lim_{p \rightarrow +\infty} -\frac{e^u}{2} \Big|_{-1}^{-p^2} \\ &= \lim_{p \rightarrow +\infty} -\frac{e^{-p^2}}{2} + \frac{e^{-1}}{2} \\ &= \frac{e^{-1}}{2} \end{aligned}$$

2. Utilice el cambio de variable  $x = \tan \theta$  para calcular  $\int \frac{1}{x^2 \sqrt{x^2 + 1}} dx$ .

5 Pts

**Solución:**

$$x = \tan \theta$$

$$dx = \sec^2 \theta d\theta$$

$$\begin{aligned} & \int \frac{1}{x^2 \sqrt{x^2 + 1}} dx \\ &= \int \frac{1}{\tan^2 \theta \sec \theta} \sec^2 \theta d\theta \\ &= \int \frac{1}{\tan^2 \theta} \sec \theta d\theta \\ &= \int \frac{1}{\frac{\cos \theta}{\sin^2 \theta}} \frac{1}{\cos \theta} d\theta \\ &= \int \frac{\sin^2 \theta}{\cos^2 \theta} d\theta \\ &= -\frac{1}{\sin \theta} \\ &= -\frac{\sqrt{x^2 + 1}}{x} \end{aligned}$$

$$x = \tan \theta \Rightarrow \sin \theta = \frac{x}{\sqrt{x^2 + 1}}$$

3. Utilice sumas de Riemann para calcular  $\int_2^4 (x + 3) dx$ .

4 Pts

**Solución:**

$$\int_2^4 (x + 3) dx = \lim_{n \rightarrow +\infty} \sum_{i=1}^n \Delta x \cdot f(x_i)$$

$$f(x) = x + 3$$

$$\Delta x = \frac{4 - 2}{n} = \frac{2}{n}$$

$$x_i = 2 + \Delta x \cdot i = 2 + \frac{2}{n} \cdot i = 2 + \frac{2i}{n}$$

$$\begin{aligned} & \sum_{i=1}^n \Delta x \cdot f(x_i) \\ &= \sum_{i=1}^n \frac{2}{n} \cdot f\left(2 + \frac{2i}{n}\right) \\ &= \sum_{i=1}^n \frac{2}{n} \cdot \left(2 + \frac{2i}{n} + 3\right) \end{aligned}$$

$$\begin{aligned}
&= \frac{2}{n} \sum_{i=1}^n \left( 5 + \frac{2i}{n} \right) \\
&= \frac{2}{n} \left( \sum_{i=1}^n 5 + \frac{2}{n} \sum_{i=1}^n i \right) \\
&= \frac{2}{n} \left( 5n + \frac{2}{n} \cdot \frac{n(n+1)}{2} \right) \\
&= 10 + \frac{2n+2}{n}
\end{aligned}$$

$$\begin{aligned}
&\int_2^4 (x+3) dx \\
&= \lim_{n \rightarrow +\infty} \left( 10 + \frac{2n+2}{n} \right) \\
&= 12
\end{aligned}$$

4. Sea  $f$  una función integrable tal que  $\int_0^x f(t) dt = \int_{x^2}^1 (t+1) dt + 3x^2 - \frac{3}{2}$ . Determine  $f(x)$ . **2 Pts**

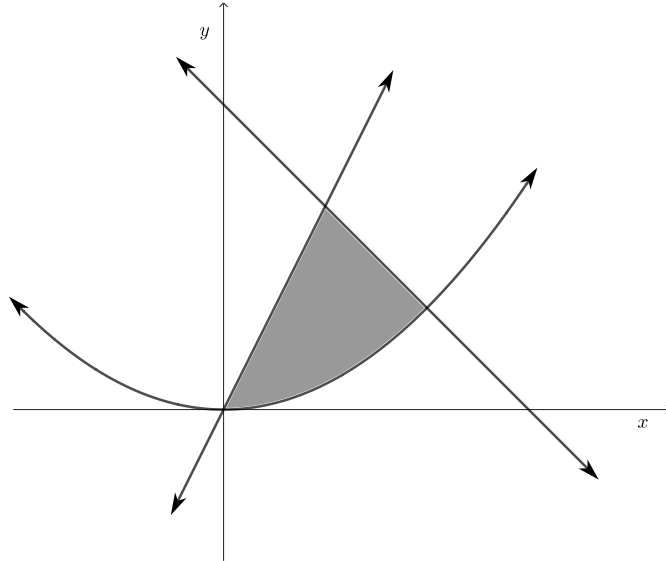
**Solución:**

$$\int_0^x f(t) dt = \int_{x^2}^1 (t+1) dt + 3x^2 - \frac{3}{2}$$

$$\Rightarrow f(x) = -(x^2 + 1) \cdot 2x + 6x$$

$$f(x) = -2x^3 + 4x$$

5. Considere la región destacada en gris, limitada por las gráficas de las funciones  $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2$ ,  $g : \mathbb{R} \rightarrow \mathbb{R}, g(x) = 4x$ ,  $h : \mathbb{R} \rightarrow \mathbb{R}, h(x) = -2x + 3$ . Plantee y calcule integrales que permitan determinar el área de la región. **5 Pts**



**Solución:**

$$4x = -2x + 3 \Rightarrow x = \frac{1}{2}$$

$$x^2 = -2x + 3 \rightarrow x = 1$$

$$A = \int_0^{1/2} (4x - x^2) dx + \int_{1/2}^1 (-2x + 3 - x^2) dx$$

$$= 2x^2 - \frac{x^3}{3} \Big|_0^{1/2} + -x^2 + 3x - \frac{x^3}{3} \Big|_{1/2}^1$$

$$= \frac{11}{24} + \frac{5}{3} - \frac{29}{24}$$

$$= \frac{11}{12}$$